Terminology and Scope

Axial Physiology Lab — Working Note APL-001 *Physiological foundation layer*

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1. Purpose

This note is the **foundation layer** for Axial Physiology Lab (APL). It sets out the basic physiology on which all later APL work rests: the tissues and systems that respond to physical loading, and the shared cellular principle through which they do so. It is deliberately general — the specific landscape of *methods* (vibration and its neighbours, movement, thermal stimulus, manual therapy) is mapped in a separate note, APL-002, and the lab's own synthesising frame (Axial Precision Vibration) in yet another. Here the subject is the **body**, not the intervention.

Vibration appears only lightly, as one of several physical stimuli. This is intentional: the physiology below is independent of the type of stimulus, and stating this plainly keeps the later, more detailed documents honest.

This is a **working note**: self-published, versioned, not peer-reviewed. It is not a clinical guideline.

2. Scope

APL studies how the human body — particularly the tissues and systems organised along the axial, load-bearing chain — responds to **physical and mechanical stimuli**. In scope at this foundational level: the shared principle of mechanobiology (§3.1) and its expression in four systems — **skeletal** (§3.2), **neuromuscular** (§3.3), **fascial / connective tissue** (§3.4), and **lymphatic** (§3.5) — bound together by a common organising frame (§3.6).

Out of scope for this note: claims of clinical efficacy; diagnostic or therapeutic advice; product marketing; and the detailed catalogue of intervention methods (that is APL-002). The lab's own frame and the question of adherence/compliance have their own documents (§6).

2.1 Note on the term "axial physiology"

"Axial physiology" is a **working scope label** for APL, not a recognised academic discipline. It denotes the study of physiological responses organised around the longitudinal (axial) loading axis of the body — the line along which gravitational, postural, and applied loads pass through the axial skeleton and the surrounding chain of soft tissue — across the skeletal, neuromuscular, fascial, and lymphatic systems. The term is adopted for clarity of scope, not as a standardised designation of a field.

3. Basic physiological concepts

APL singles out four systems — skeletal, neuromuscular, fascial, and lymphatic — not because they exhaust the physiology of the response to mechanical loading. Other systems are mechanosensitive too (among them the vascular, cutaneous, and respiratory systems). The selection reflects APL's scope: these are the principal tissues of the axial, load-bearing chain for which well-characterised mechanobiological mechanisms exist. The remaining systems either enter the response secondarily or lie off the load-bearing axis — and so are not developed here.

3.1 Mechanobiology and mechanotransduction (the unifying principle)

Living tissue adapts to the mechanical loads placed on it. **Mechanical loading** takes several forms — compression, tension, shear, bending, torsion, and oscillatory variation of these quantities. Tissues differ in their sensitivity to the type of load, its amplitude, frequency, and duration. Vibration is one of these forms — oscillatory loading — not a separate physiological category.

This adaptation serves **homeostasis**, not the "building" of a tissue: the organism continually matches structure and function to a changing mechanical environment so as to maintain capacity under a given load [10]. Growth (of bone, of muscle) is one possible direction of this regulation, not its goal in itself.

The shared cellular basis is a pair of processes. First **mechanosensing**: the cell detects a mechanical stimulus — deformation, fluid movement, shear stress, stretch. Then **mechanotransduction**: the conversion of that stimulus into a biochemical response that changes the cell's behaviour [1]. These forces reach cells largely through the **extracellular matrix (ECM)** — the network that transmits and distributes load within the tissue — as well as through cell–cell junctions and the cytoskeleton; the ECM is therefore one of the principal media of transmission, not the only one [1,10]. **Mechanobiology** is the broader study of this active coupling between mechanical signals and biological pathways.

These responses are not linear. They are **dose-dependent and threshold-gated** (below a certain stimulus there is no response; above another the response may saturate or reverse), **time-dependent**, and modulated by the organism's **biological state** — age, hormonal status, sleep, nutrition, inflammation. The same stimulus therefore does not produce the same response in everyone or under all conditions. This is one reason the existence of a mechanism does not translate directly into a predictable effect (§5).

Each of the systems below is an expression of the same principle in a different tissue; for this reason a single physical stimulus can, in principle, reach several systems at once.

3.2 Skeletal system

Bone adaptation. Bone remodels in response to mechanical loading — a regularity summarised as **Wolff's law**, with mechanotransduction as its cellular description [1]. The **osteocyte** (~90–95% of bone cells) is the principal mechanosensor, detecting load mainly through **fluid shear stress** in the lacunar–canalicular network [1,2]. Mechanical signals can be **anabolic** for bone (favouring formation over resorption) even at low amplitude [3].

3.3 Neuromuscular system

Proprioception is the sense of body position and movement, served by receptors in muscles, tendons, and joints; it feeds **sensorimotor integration**, the nervous system's combining of proprioceptive, visual, and vestibular signals into coordinated posture and movement [4]. The **muscle spindle**, whose primary afferents (Ia) respond strongly to mechanical oscillation, is the main channel through which mechanical stimulus enters this system [4]. It is not the only one — cutaneous, fascial, and joint mechanoreceptors also respond to stimulus — but the spindle is the principal entry point for oscillation, and those other pathways are the subject of APL-002. This is the pathway on which any effect on balance, postural control, or motor activity would manifest.

3.4 Fascial / connective-tissue system

Fascia is a body-wide, layered, multiscale connective-tissue network that enables tensional loading and shearing mobility along its interfaces; recent work proposes treating it as an anatomical **system** composed of four organs — superficial, musculoskeletal (deep), visceral, and neural fascia [5]. This proposal is the subject of an ongoing nomenclatural debate and has met published criticism [9]; APL adopts it as a useful descriptive frame, not as settled consensus. Fascia is not passive filler: it is richly innervated and has a sensory role. Gliding between fascial layers is enabled by **hyaluronan (HA)** in the loose connective-tissue interfaces, secreted by **fasciacytes** (a fascia-specific cell type described by Stecco's group) [5,6]. Under overload, hypoxia, or inactivity, HA can aggregate — a process described in Stecco's framework as **densification** — which is held to increase stiffness, impair gliding, and sensitise nociceptors [6]. This mechanism is proposed and increasingly accepted, but not entirely uncontested. Because fascia is continuous and load-bearing, it is a plausible route by which mechanical stimulus spreads beyond the point of application.

3.5 Lymphatic system

The **lymphatic system** returns interstitial fluid to the circulation and transports immune cells. Lymph is moved by two pumps: an **extrinsic** one (compression of vessels by surrounding tissues and skeletal-muscle movement) and an **intrinsic** one (rhythmic, phasic contraction of lymphatic muscle in the collecting vessels) [7,8]. Crucially, the lymphatic pump is **mechanosensitive**: a rise in pressure/stretch typically activates intrinsic contraction, while flow-induced shear stress modulates it — enhancing it or, above a threshold, inhibiting it [7, 8]. This makes the lymphatic system in principle sensitive to external mechanical and rhythmic stimulus — a mechanistic basis worth stating plainly and, equally, worth not overstating.

3.6 The mechanobiological axis (organising frame)

The four systems above are the same principle in different tissues. APL adopts their shared course as an **organising frame** — not a research model or a claim of discovery, but a naming scaffold that the later notes (APL-002, APV) refer back to:

> mechanical stimulus → force transmission through ECM and tissues → mechanosensing → mechanotransduction → cellular response → tissue adaptation → change in system function → organism-level effect

The frame runs across increasing scales of organisation, from a single force to the response of the whole organism. As one moves from the cellular level to the organism level, the number of factors bearing on the outcome grows, so that the link between a cellular mechanism and the effect observed in a human becomes progressively less direct. A separate dimension — **evidence distance** (how close to the human a given measurement was made) — is developed in §5. The frame is common to all classes of physical stimulus in §4; it is not a property of any single method.

4. Physical stimuli (light accent)

The physiology above is **independent of the type of stimulus**: the same systems are reached by several classes of physical stimulus — mechanical **loading and movement** (exercise), **thermal** stimulus, **manual therapy and massage**, and mechanical **vibration**. APL's target interest is low-intensity vibration, but that is a choice of method, not a property of the physiology. The full landscape of methods — how each is defined, its parameters, its evidence — is the subject of **APL-002**, not this note.

5. Mechanism versus efficacy: the evidence-distance axis

Most of the above is established at the tissue and cell level: bone, muscle spindle, fascia, and lymphatic vessel — each demonstrably responds to mechanical signals. That a mechanism exists does **not** prove that a given intervention yields a clinically meaningful outcome.

This distinction is worth grounding in an explicit **evidence-distance axis** — from evidence furthest from the human to closest:

> in vitro (cell in culture) → ex vivo (isolated tissue) → in vivo animal → human: mechanism → human: clinical outcome

The nearer the left end, the cleaner the control over the mechanism — and the greater the distance to the patient. **In vitro** data are strongest for *showing that a mechanism operates* and weakest for *claiming a clinical effect*. A result obtained in cells in a dish is not confirmation of efficacy in a human; it is a mechanistic premise that still has to be carried across the successive links of the axis. This boundary is easy to cross in communication — a result from culture is sometimes dragged straight into a promise to the patient — and APL keeps it explicit precisely for that reason.

APL maintains this separation throughout: mechanism is described in §3; the efficacy of specific methods is treated cautiously and case by case in later documents, where the evidence is often ambiguous, and where each method's position on the axis above is noted explicitly.

6. Related APL documents

This note is the foundation layer. It connects to:

- **Method landscape (APL-002)** — the broad catalogue of physical interventions (the vibration family: LIV, LMHF/LMHFV, WBV, vibroacoustics, vibrotactile stimulation, local vibration, entrainment; plus movement, thermal stimulus, manual therapy), each defined with its parameters and evidence status.
- **Axial Precision Vibration (APV)** — the lab's own *synthesising frame* (a seated-first synthesis prioritising adherence), in its own note, plainly marked as a lab frame rather than an established modality, and positioned so as not to inherit the evidence base of the methods it draws on.

- **Adherence / compliance** — a dedicated article, including a proposed seated-vs-platform comparison (note: a platform-based seated LIV protocol for older wheelchair users already exists in the literature and must be cited and distinguished).
- **Research agenda / open questions (APL-005)** — directions the lab intends to take, including questions that cut across individual systems (e.g. mechano-metabolic coupling [11]). A statement of intent, not of results.

7. Citation and versioning

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8. Disclosure

The author is the founder of Longevity Space — a commercial entity that develops low-intensity vibration devices (including AEVUM). Axial Physiology Lab is an independent research undertaking affiliated with Longevity Space. This work was self-funded and received no external research funding. This disclosure allows any reader to take the author's commercial interest into account when reading APL materials.

References

1. Duncan RL, Turner CH. Mechanotransduction and the functional response of bone to mechanical strain. *Calcif Tissue Int*. 1995;57(5):344–358. doi:10.1007/BF00302070

2. Fritton SP, McLeod KJ, Rubin CT. Quantifying the strain history of bone: spatial uniformity and self-similarity of low-magnitude strains. *J Biomech*. 2000;33(3):317–325. doi:10.1016/S0021-9290(99)00210-9

3. Rubin C, Turner AS, Bain S, Mallinckrodt C, McLeod K. Anabolism: low mechanical signals strengthen long bones. *Nature*. 2001;412(6847):603–604. doi:10.1038/35088122

4. Amiez N, Beaulieu L-D, Paizis C, Lapole T, Di Rienzo F. Muscle tendon vibration revisited: why tonic vibration reflex and kinaesthetic illusions matter. *J Physiol*. 2026;604(8):3231–3252. doi:10.1113/JP290085

5. Stecco C, Pratt R, Nemetz LD, Schleip R, Stecco A, Theise ND. Towards a comprehensive definition of the human fascial system. *J Anat*. 2025;246(6):1084–1098. doi:10.1111/joa.14212

6. Stecco C, Stern R, Porzionato A, Macchi V, Masiero S, Stecco A, De Caro R. Hyaluronan within fascia in the etiology of myofascial pain. *Surg Radiol Anat*. 2011;33(10):891–896.

7. Munn LL. Mechanobiology of lymphatic contractions. *Semin Cell Dev Biol*. 2015;38:67–74. doi:10.1016/j.semcdb.2015.01.010

8. Scallan JP, Zawieja SD, Castorena-Gonzalez JA, Davis MJ. Lymphatic pumping: mechanics, mechanisms and malfunction. *J Physiol*. 2016;594(20):5749–5768.

9. Scarr G. Response to the recent article: Stecco et al. (2025) "Towards a comprehensive definition of the human fascial system." *J Anat*. 2025;247:408–409. doi:10.1111/joa.14258

10. Humphrey JD, Dufresne ER, Schwartz MA. Mechanotransduction and extracellular matrix homeostasis. *Nat Rev Mol Cell Biol*. 2014;15(12):802–812. doi:10.1038/nrm3896

11. Romani P, Valcarcel-Jimenez L, Frezza C, Dupont S. Crosstalk between mechanotransduction and metabolism. *Nat Rev Mol Cell Biol*. 2021;22(1):22–38. doi:10.1038/s41580-020-00306-w

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